Embedded Systems Development

Exercise 1

1. Construct a microcontroller-based system to chase five LEDs from left to right repeatedly.

   a. If the system is supplied with 5 V and an LED is driven with 2 V and 15 mA, what is the resistor required to limit the current to the LED? Show your workings clearly.

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   b. Prepare a Bill of Materials (BOM).

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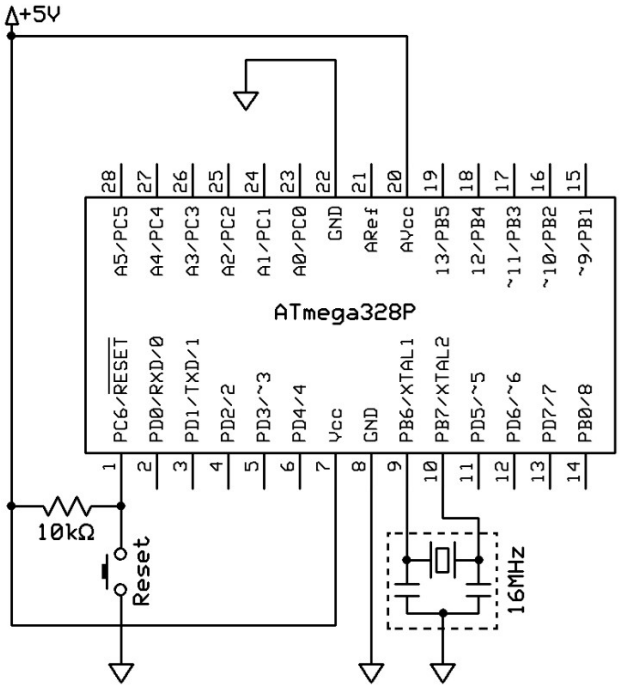
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   c. Complete the following schematic diagram for your hardware design.



d. Write the required firmware using the Arduino wrapper functions.

e. Write the required firmware in C.

2. Modify your system to drive the five LEDs in Knight Rider pattern.
   a. Write the required firmware using the Arduino wrapper functions.

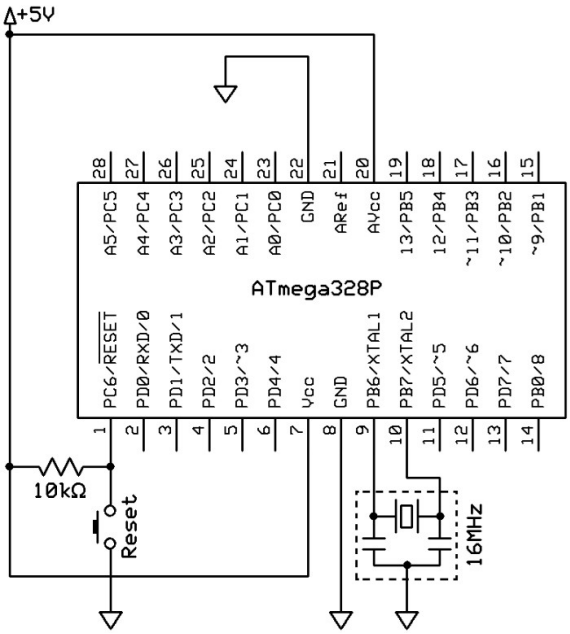
b. Write the required firmware in C.

3. Extend the system developed in Question 2 to drive 10 LEDs using a maximum of seven I/O pins.

   Hint: Use two BC547 transistors with 1 kΩ base resistors to switch the GND connection between the cathodes of the first and second sets of five LEDs.



   a. Complete the following schematic diagram for your hardware design.



b. Write the required firmware in C.